

Non-ergodic Mess3 Mixture Experiment

Eshika Pathak

Experiment report

This report studies a non-ergodic mixture of three Mess3 processes. For each training sequence, one ergodic component is sampled once and then kept fixed for the whole sequence, so the global hidden process is a block-diagonal 9-state hidden Markov model. A small decoder-only transformer is trained by next-token prediction on sequences of length 20 over a 4-token vocabulary made up of three observation symbols plus BOS.

1. Dataset construction and training setup

I used three Mess3 components with parameters $M1 = (\alpha = 0.90, x = 0.05)$, $M2 = (\alpha = 0.55, x = 0.20)$, and $M3 = (\alpha = 0.25, x = 0.05)$. Each training sequence is generated entirely by one of these components, chosen once at the start of the sequence. This makes the overall process non-ergodic, because the sequence never switches between components after generation begins. The transformer is then trained on this dataset with standard next-token prediction.

quantity	value
training steps	5000
batch size	128
model	2 layers, d_model=24
final train loss	0.9814
training time (s)	130.1
final-layer final-position R^2 (component posterior)	0.952
final-layer final-position R^2 (full 9-state belief)	0.645
avg true component entropy pos 1	1.099
avg true component entropy pos 20	0.571

2. Why this structure is interesting for language models

I think this structure is interesting because real language data are not generated by one single stationary source. Different users, domains, writing styles, or tasks can behave like different latent generators. A non-ergodic mixture is a simple way to model that. Within one context window, the model may need to figure out which generator is active and then adapt its predictions accordingly. So this experiment acts like a toy version of in-context adaptation: first identify the active component, then track where you are inside that component.

3. Honor-code pre-registered prediction

Before looking at the trained-model plots, my prediction was that the residual stream would develop a two-scale geometry. At early context positions, activations from different components should overlap more because the model is still uncertain about which component generated the sequence. As more context is observed, the representation should separate more clearly by component. Within high-confidence regions for a given component, I expected the activations to reflect the local 3-state belief geometry of that component. I also expected later layers to show cleaner and more linearly readable geometry than earlier layers.

4. Mathematical derivation of the predicted geometry

Let c denote the ergodic component and let s denote the local hidden state within that component. Since there are 3 components and each component has 3 hidden states, the full hidden state can be written as the pair (c, s) , giving 9 total hidden states.

The key structural fact is that the process never jumps between components once a sequence starts. So the global transition operator is block diagonal: each component evolves inside its own 3-state block, and there are no cross-component transitions.

If $b_t(c, s)$ is the posterior probability of being in component c and local state s after observing the first t tokens, then a Bayesian update at the next step only mixes over states inside the same component. That means the posterior naturally splits into two pieces: a posterior over which component generated the sequence, and a conditional posterior over the local state inside that component.

In other words, the 9-state belief can be written in the form $b_t(c, s) = q_t(c) \times a_{t,c}(s)$, where $q_t(c)$ is the 3-way posterior over components and $a_{t,c}(s)$ is the local 3-state belief inside component c .

This factorization gives the geometric prediction. The coarse part of the belief, $q_t(c)$, lives in a 3-way simplex, so globally we should see a component-level simplex geometry. Then, once one component has high posterior, the remaining uncertainty is only over the 3 local states inside that component, so within each branch we should see a local 3-state simplex geometry. That is why I expected a mixture-of-geometries structure rather than one single global shape.

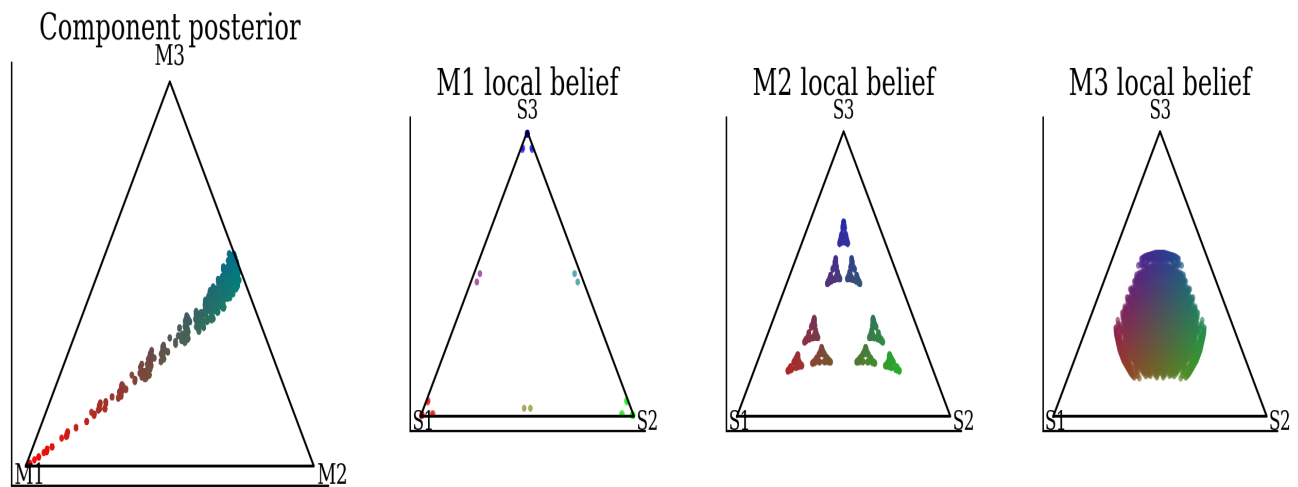
5. Geometric intuition

The simplest way I think about it is: first the model is trying to answer “which world am I in?”, and after that becomes clearer, it is trying to answer “where am I inside that world?” So early in the sequence I expect activations to sit closer to an overlap region, because the component is still ambiguous. Later in the sequence I expect the representation to split apart by component. Within each branch, I expect a smaller local geometry corresponding to the hidden-state uncertainty of that component.

6. Multiple possible geometries

There were at least three plausible possibilities before training. The first was the ideal two-scale geometry: activations separate by component posterior and also retain the local 3-state belief geometry within each branch. The second was a coarser geometry: later layers might mainly encode only the component posterior, so the representation would collapse into three broad clusters with little visible within-component structure. The third was a noisier or curved version of the two-scale picture, where component identity is clear but the local geometry only survives approximately rather than as a clean simplex-like object.

7. Predicted geometry

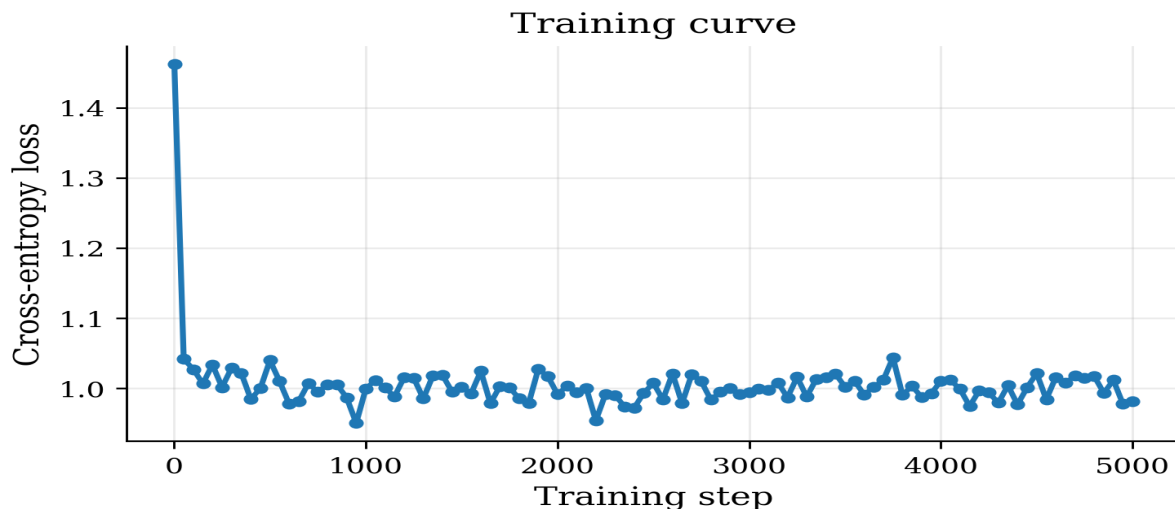


What is plotted. The left panel shows the exact Bayesian posterior over the three ergodic components, mapped into a 2D probability simplex. Each point is one observed prefix. The triangle vertices represent certainty in one component: bottom-left = M1, bottom-right = M2, and top = M3. Point color also shows the component posterior itself: red, green, and blue intensities indicate posterior mass on M1, M2, and M3, respectively. The three panels on the right show the local 3-state belief geometry inside each component, after restricting to prefixes where that component has high posterior. In those panels, the triangle vertices represent certainty in the three local hidden states of that component: bottom-left = S1, bottom-right = S2, top = S3. Point color again gives the 3-way local belief vector, so mixed colors indicate uncertainty across local states.

Why this matters. This is the exact geometry predicted by Bayesian filtering, so it is the main target structure against which the transformer's residual-stream geometry can be compared.

What we expect. A two-scale structure: globally, the geometry should organize by component posterior, and locally, each component branch should contain its own 3-state belief geometry.

8. Optimization



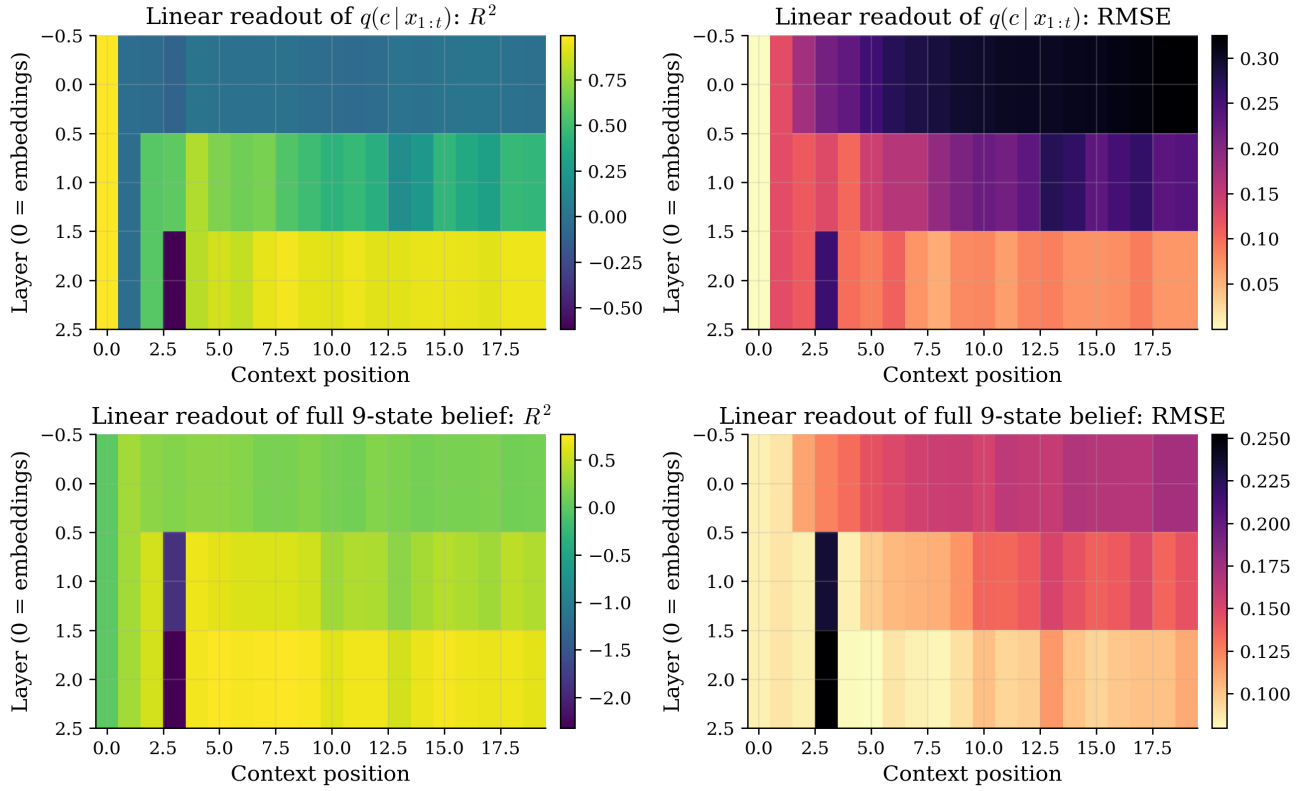
What is plotted. Training cross-entropy loss as a function of optimization step.

Why this matters. Before interpreting geometry, I wanted to check that the model was actually learning predictive structure from the data.

What we expect. If the model is using context to infer component identity and latent state, then the loss should fall below the

uninformed baseline.

9. Linear readout from the residual stream

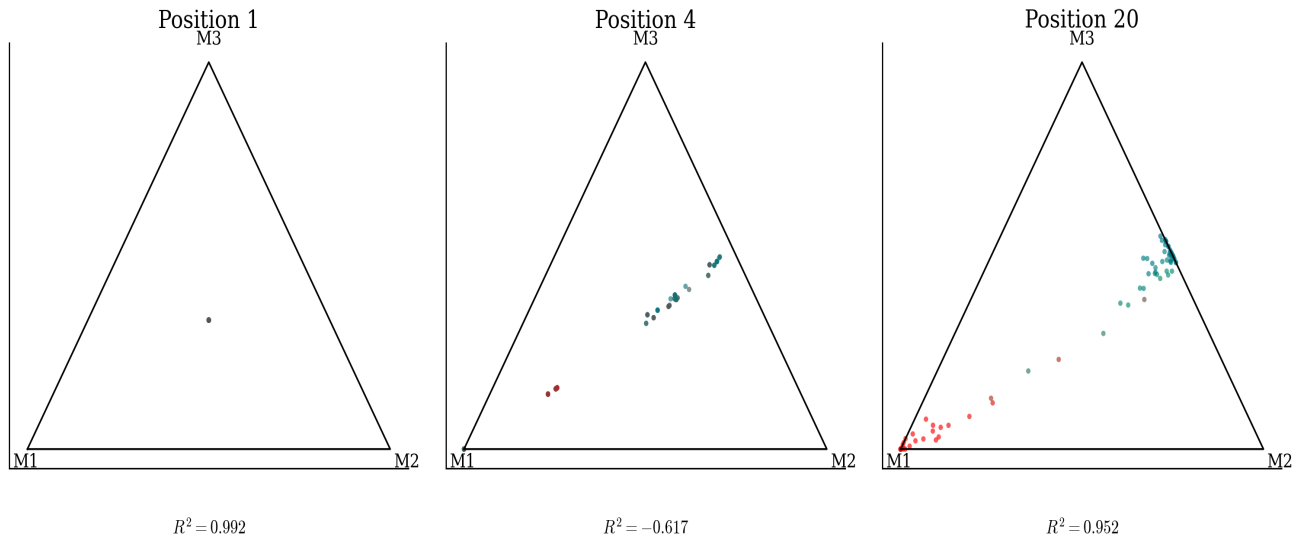


What is plotted. Each heatmap measures how well a linear probe can recover a target belief quantity from the residual stream at a particular layer and context position. The top row is for the 3-way component posterior and the bottom row is for the full 9-state hidden-state belief. Higher R^2 and lower RMSE mean that the corresponding latent variable is more linearly readable from the representation.

Why this matters. These plots show where in the network and where in the sequence the model is carrying coarse component information versus finer hidden-state information.

What we expect. Component identity should become readable earlier and more strongly than the full 9-state belief, and both should generally improve at deeper layers and later positions.

10. Learned component geometry

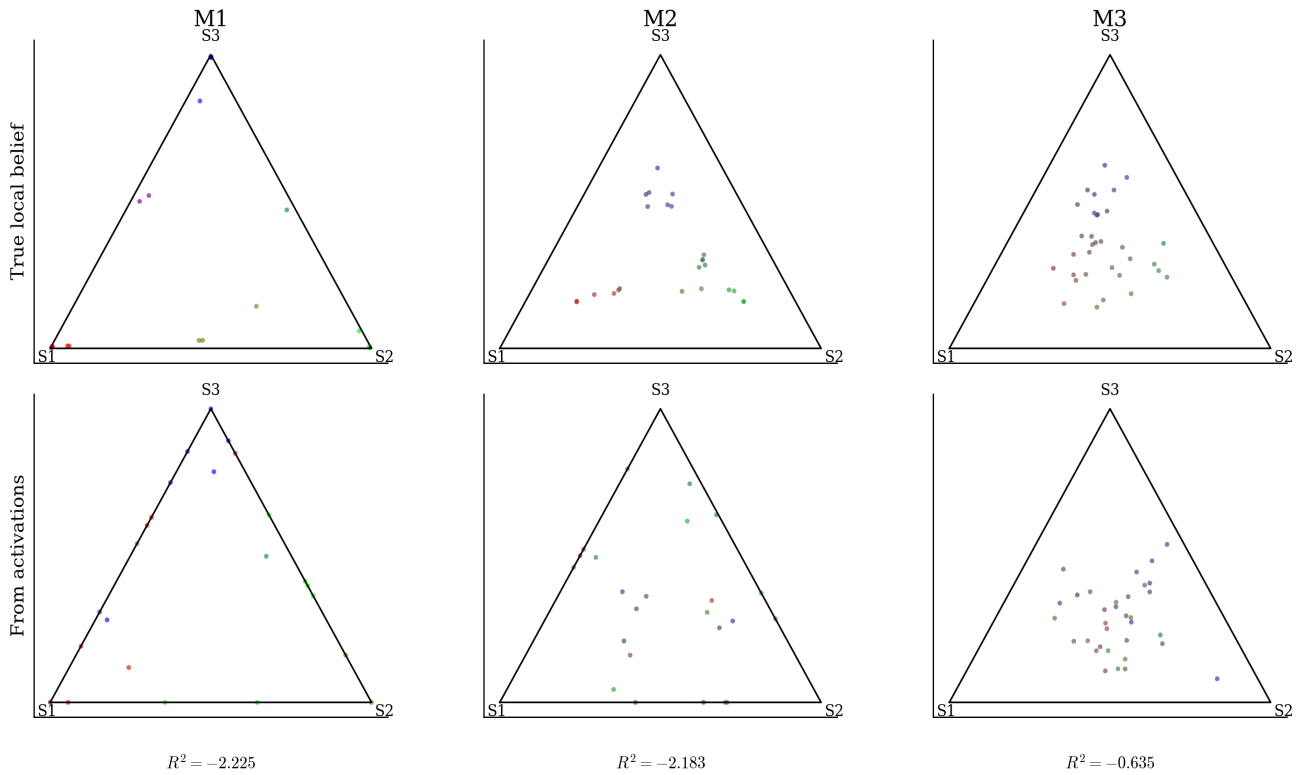


What is plotted. Final-layer residual activations at selected context positions are linearly projected into the best-fit 3-way simplex for the component posterior. Each point is one held-out example. The triangle vertices represent certainty in M1, M2, and M3; specifically bottom-left = M1, bottom-right = M2, and top = M3. Point color shows the true component posterior for that example, using the same RGB convention as before: the color channels indicate posterior mass on M1, M2, and M3.

Why this matters. This gives a direct visual comparison between the network's learned component-level geometry and the exact Bayesian geometry.

What we expect. Early in the sequence, points should cluster closer to the uncertain interior of the simplex. As more context is observed, points should move toward component-specific regions, reflecting better synchronization to the correct ergodic component.

11. Relation to the component processes



What is plotted. Each column corresponds to one Mess3 component. The top row shows the true local 3-state belief geometry within that component, after conditioning on examples where the component posterior is high. The bottom row shows the corresponding geometry obtained by linearly projecting final-layer activations. In each panel, the triangle vertices represent certainty in one of the three local hidden states for that component: bottom-left = S1, bottom-right = S2, top = S3. Point color shows the true local 3-state belief vector, so mixed colors indicate uncertainty across local states.

Why this matters. This tells us whether the network has learned only coarse component classification, or whether it also captures the finer within-component latent-state structure.

What we expect. If the learned representation really has the predicted two-scale belief geometry, then the bottom-row projected activations should resemble the top-row true local belief geometry.

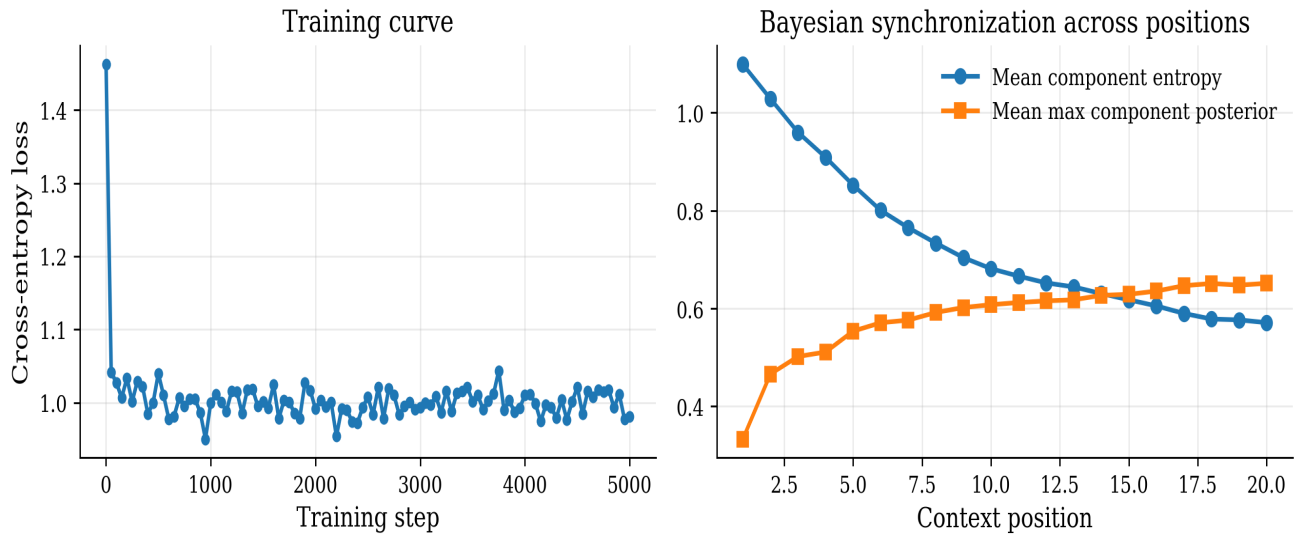
12. Direct answer: what structure is present after training?

After training, the clearest structure in the residual stream is the posterior over ergodic component. Later positions and deeper layers show stronger component separation than early positions and shallow layers. Within examples where the model is already fairly certain about the component, the residual stream also retains information about the local 3-state belief of that component, although this finer structure is weaker than the coarse component geometry. So the learned representation is best viewed not as one single simplex, but as a mixture of a component-level geometry together with local within-component belief geometry.

13. Additional analysis of my choosing

As an additional analysis, I looked at Bayesian synchronization across context position through the mean component entropy and the mean maximum component posterior. I chose this because it helps separate the difficulty of the statistical problem from the behavior of the transformer itself. If the exact Bayesian posterior becomes sharper with more context, then later component separation in the network is something we should expect from the data-generating process itself.

14. Additional analysis: Bayesian synchronization across position



What is plotted. The left panel shows training loss over optimization. The right panel shows two summary statistics of the exact Bayesian posterior over components as a function of context position: mean component entropy and mean maximum component posterior.

Why I chose this analysis. This helps separate two questions: whether the transformer is learning useful structure, and whether the underlying inference problem itself becomes easier as more tokens are observed.

What we expect. As context grows, the exact posterior over components should become sharper. So mean entropy should decrease and the mean maximum component posterior should increase. That would support the interpretation that later component separation in the residual stream is statistically natural for this data-generating process.

Another analysis I think would be very informative, even though I did not implement it here, would be to compare the transformer's next-token cross-entropy to the exact Bayesian next-token predictor at each context position. That would tell us how close the learned model is to the Bayes-optimal predictor for this non-ergodic hidden Markov model.

15. Main takeaways

- A non-ergodic Mess3 mixture creates a natural two-scale latent inference problem: identify the active component, then track the local state within it.
- The mathematically predicted belief factorization naturally leads to a two-scale geometry in the residual stream.
- Empirically, component posterior is the strongest and most linearly accessible structure; finer within-component belief structure is present but weaker.
- This makes the experiment a useful toy model of in-context adaptation in language models, where the model must infer which latent generator is currently active.

Caveat: this is still a small proof-of-concept on CPU, so it should be read as an implementation and qualitative geometry study because my model training isn't good enough to draw strong conclusions.